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Think Academy

Round 2

Onsite 11+ Boost for SPG/Westminster & Top Independents



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Name: _____

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Lesson 1

Fractions

Exploration 1

One kilometre is approximately $\frac{5}{8}$ of a mile.

One gallon is approximately $4\frac{1}{2}$ litres.

A car travels $26\frac{2}{5}$ kilometres. The car uses up $2\frac{1}{5}$ litres of petrol during the journey.

Using the conversion factors above:

a. How far did the car travel in miles?

b. How many gallons of petrol did it use?

c. What was its average petrol consumption in **miles per gallon**?
Give your answers as mixed numbers.

Exploration 2

Mr Nomad is planning a family holiday in Manchuria.
He will be going with his wife and his three children, Matthew, aged 10, Mark, aged 8 and Luke, aged 2.
There are three main costs for him to consider – the flights, the villa they will stay in, and car hire. He collects the following information.

Currency

In Manchuria, the unit of currency is a wot and £1 is worth 15 wots.

Flight costs

£350 per person including all airport taxes and fuel surcharges.

Children travel at full fare except that children under 5 travel at $\frac{2}{5}$ of the full fare.

Villa

£30 per person per night.
Children under 3 are free and there is a 40% reduction for children under 12.

Car hire

180 wots per day and you supply your own fuel.
Fuel costs 18 wots per litre and the car travels 6 miles per litre.

Mr Nomad is planning a holiday for 8 nights in the villa. He will hire a car for 5 days and will travel an average of 48 miles per day. Work out the total holiday cost and give your answer in pounds.

Exploration 3

1 Sarah has a large number of marbles. She gives two fifths of her marbles to her friend, and then gives 18 marbles to each of her three brothers. She then has 48 marbles left. How many marbles did she originally have?

2 Lucy also has a large number of marbles. She gives $\frac{3}{7}$ of her marbles to her friend Alex, and $\frac{2}{9}$ of her marbles to her other friend Rupert. Alex receives 39 more marbles than Rupert. How many marbles did Lucy originally have?

Exploration 4

1 Vivien eats one tenth of a chocolate cake. Allan then eats one sixth of the remaining cake. Bella finishes the cake. What fraction of the cake does Bella eat?

2 On Monday, I ate one sixth of a cake.
On Tuesday, I ate one fifth of the remaining cake.
On Wednesday, I ate one quarter of the remaining cake.
On Thursday, I ate one third of the remaining cake.
On Friday, I ate half of the remaining cake.
What fraction of the cake remained at the end of the week?

Exploration 5

Simon weighs 54 kg plus a third of Robert's weight.
Robert's weight is three quarters of Simon's weight.
What is Simon's weight in kg?

Homework

1 A baker is following a recipe. He knows the following conversions:

One cup is approximately $\frac{2}{5}$ of a litre.

One gram is approximately $\frac{3}{85}$ of an ounce.

The recipe requires $4\frac{1}{2}$ cups of milk and $5\frac{2}{5}$ ounces of butter. The baker makes a batch of cakes that uses up the full amount of these ingredients.

a. How much milk did the baker use in litres? _____ litres (in decimal)

b. How much butter did the baker use in grams? _____ grams

c. How many grams of butter are needed for per litre of milk? _____ grams

2 The Smith family are planning a camping trip to Spain. The family consists of Mr and Mrs Smith, and their two children, Emily, aged 10, and Ben, aged 3. They are planning a 3-night trip.

£1 = 1.2 euros.

Flights: £280 per adult, children under 5 pay $\frac{2}{5}$ fare.

Apartment: €50 per person per night. Children under 4 are free, under 12 get 30% off.

Car hire: €45 per day, fuel €1.80 per litre, car travels 14 km per litre. They hire a car for 3 days, and drive 70 km per day.

Find the total cost in pounds. (Give your answer in decimal.)

Answer: £ _____

3 In a school election for head student, $\frac{4}{9}$ of the students voted for Chloe and $\frac{4}{15}$ of the students voted for Thomas. Chloe received 40 more votes than Thomas. How many students voted in the election in total? _____

4 On Sunday, Aisha reads $\frac{1}{10}$ of her book.

On Monday, she reads $\frac{1}{9}$ of the remaining pages.

On Tuesday, she reads $\frac{1}{8}$ of the remaining pages.

On Wednesday, she reads $\frac{1}{7}$ of the remaining pages.

What fraction of the book is still unread? _____

5 The length of a rectangular garden is 12 metres more than half of its width.
The width of the garden is exactly two-fifths of its length.
What is the length of the garden in metres? _____ m

Lesson 2

Ratios

Exploration 1

A mother and her two children, Lily and Harry, have just won the lottery. They have decided to share their fortune according to the ratio of their ages.
Mum is 40. Lily is two years older than Harry, and Mum is two years older than their combined age.
Mum decides to give her share to a dogs' home and a cats' home in the ratio 7:9.
Lily and the dog's home receive £15 million between them. How much more money did the cats' home receive than Harry?

Exploration 2

A school is ordering decorations for the Christmas party.
For every garland, there are 2 stars. For every 3 stars, there are 4 pinecones, and for every 5 pinecones, there are 6 baubles.
There are 532 items ordered in total. How many baubles were ordered?

Exploration 3

1 Damon and Arthur went to a BBQ. They ordered some drinks.
Arthur had 2:3 orange to lemonade juice and
Damon had 1:5 grapes to lemonade juice.
Arthur drank double the amount of Damon.
What fraction of all the lemonade did Arthur drink?

2 The following are instructions to mix paint of various different colours (the primary colours are red, blue and yellow)
Purple: mix 2 parts red and 3 parts blue
Green: mix 1 part blue and 1 part yellow
Cyan: mix 1 part green and 1 part blue
Brown: mix 2 parts purple and 1 part green
In what ratio should blue, red and yellow paint be mixed to make brown paint?

Exploration 4

Standard green paint is a mixture of 40% blue paint and 60% yellow paint.
Dark green paint is a mixture of 50% blue paint and 50% yellow paint.
Light green paint is a mixture of 30% blue paint and 70% yellow paint.

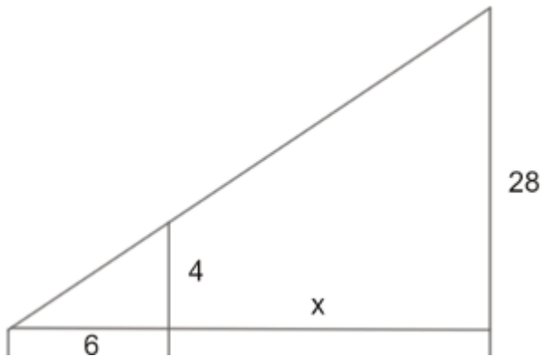
(1) I have 4 litres of standard green paint. How much blue paint do I need to add to this to make dark green paint?

(2) I have 3 litres of standard green paint. How much yellow paint do I need to add to this to make light green paint?

(3) I have 4 litres of light green paint. How much dark green paint should I add to this to obtain a mixture of 48% blue and 52% yellow paint?

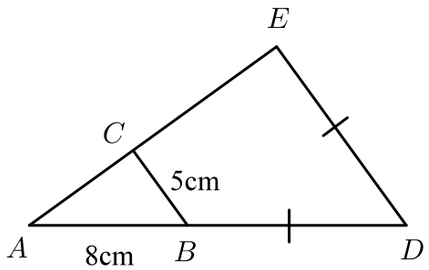
Warm Up

I am sitting 6m away from a wall of height 5m. My eye level is 1m above the ground. I notice that I can just see the top of a building, which I know to be 29m tall, some distance away on the other side of the wall.
What is the distance between the wall and the building?



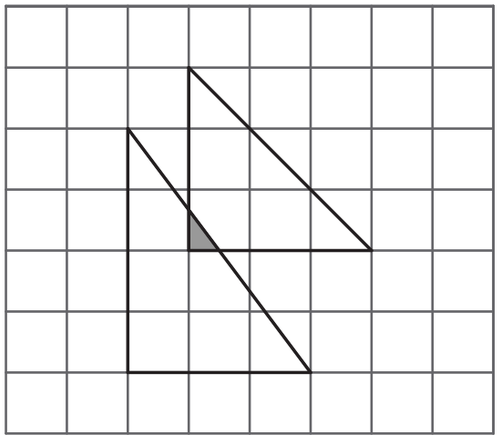
Exploration 5

1 In the diagram below, triangles ABC and ADE are similar, and lengths BD and ED are equal.
Find the length BD, giving your answer as a mixed number.



2 The diagram below shows two overlapping right-angled triangles. Work out the exact area of the shaded triangle, giving your answer as a fraction.

You must calculate the exact fraction. Estimating, measuring or guessing will gain no marks.



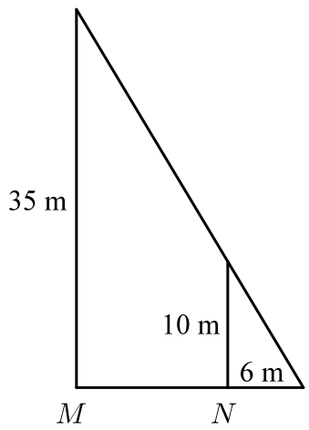
Homework

1 Four children inherit money from their father's will. They receive money in the ratio 6:5:4:3 (from eldest to youngest). The eldest two children between them receive £2500 more than the youngest two children. How much money do the four children receive in total? £ _____

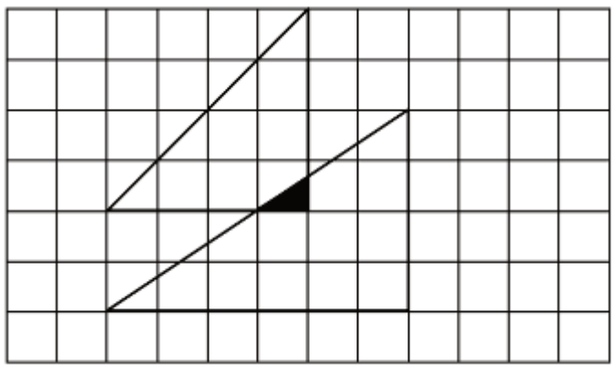
2 Harriet and Georgia race over 100m. Georgia is 10m behind when Harriet crosses the finish line.
Georgia and Indigo have a race over 100m. Indigo is 12m behind when Georgia crosses the finish line.
If Harriet and Indigo were to race over 100m, how far behind would Indigo be when Harriet crosses the finish line? _____ m
(Assume each girl runs at a constant speed. Give your answer as a decimal.)

3 Charlotte and Sophie attended a summer garden party and ordered fruit punches.
Charlotte had a punch with raspberry and lemonade in the ratio 2:5.
Sophie had a punch with peach and lemonade in the ratio 1:4.
Charlotte drank three times the amount that Sophie drank.
What fraction of all the lemonade did Charlotte drink? _____

4 The two triangles in the following figure are both right-angled triangles. Find the length of MN. _____ m



5 The diagram below shows two overlapping right-angled triangles. Work out the exact area of the shaded triangle, giving your answer as a fraction. _____



Lesson 3

Algebra

Exploration 1

1 Seb and Bill each have a collection of marbles. Initially, Seb has seven more marbles than Bill.
Over the next year, Seb doubles the number of marbles he has, and Bill triples the number of marbles he has. At the end of the year, Bill has seven more marbles than Seb.
How many marbles did they each have to start with?

2 In eight years' time, Andy will be twice as old as Beth. Ten years ago, Andy was five times as old as Beth. How old are Andy and Beth today?

Exploration 2

1 One third of the students in year 6 are members of the football club. When five more pupils in that year join the football club, two fifths of year 6 belong to the club. How many students in total are there in year 6?

2 Three eighths of year 5 belong to the football club. When six new pupils join year 5, four of them decide to join the football club. Then, two fifths of year 5 belong to the group. How many students are there in Year 5 now?

Exploration 3

1 Each Saturday, Homer buys some of his favourite pies from his local pie shop.
He doesn't remember what they each cost, but he knows that last Saturday, he bought 3 medium pies and 2 large pies, and spent a total £35.60. The previous Saturday, he bought 4 medium pies and 3 large pies, and spent a total of £50.30. How much does a medium pie cost?

2 On his stall, Nathan is selling sunhats and umbrellas. At the start of the day, he has 11 sunhats and 13 umbrellas to sell.
In the morning, it is sunny, and he charges £5 for an umbrella, and £6 for a sunhat. He makes £63 in the morning.
In the afternoon, it is drizzling, and he charges £6 for an umbrella and £5 for a sunhat. He sells all his remaining stock, and makes £64 in the afternoon.
How many umbrellas did he sell in the morning?

Exploration 4

1 Class 5A and Class 5B went to the library twice, and totally borrowed 29 books. On the first visit, 5A borrowed 6 times as many books as 5B. On the second visit, 5B borrowed 4 times as many books as 5A. How many books did Class 5B borrow from the library?

2 In the online role-playing game Empire Quest, players amass gold coins and spend them in the marketplace on either spells, potions or maps. Each of these items costs a whole number of gold coins.
Lucas buys 1 spell, 10 potions and 7 maps, spending 77 gold coins. Majid buys 1 spell, 3 potions and 4 maps, spending 41 gold coins.
Determine the price of each item.

Homework

1 Three consecutive numbers are such that five times the middle number is 60 more than the sum of the largest number and three times the smallest number. What is the middle number of the three numbers? _____

2 In a library's 'New Arrivals' section, three sevenths of the books are fiction. The librarian adds 14 more books to the section. Of these new additions, 10 are fiction books. Now, one half of the books in the section are fiction. How many books were in the section originally? _____

3 At a stationery shop, Aisha buys 5 notebooks and 4 pens for £15.60. Her friend Ben buys 6 notebooks and 5 pens for £18.90. How much does a single notebook cost? £ _____

4 A farmer at a market has 30 bags of potatoes and 25 bags of carrots.
Before lunch, she sells potatoes for £3 a bag and carrots for £2 a bag, taking in £70.
After lunch, she sells potatoes for £2 a bag and carrots for £3 a bag. She sells all her remaining stock for £85.
How many bags of carrots did she sell before lunch? _____

5 Sheila bought some grapefruits, mangoes and pineapples. Mangoes cost £1 each, grapefruits 70p each, and pineapples 90p each. She spent exactly £20 and bought 10 grapefruits. How many mangoes did she buy? _____

Lesson 4

Logic Puzzles

Exploration 1

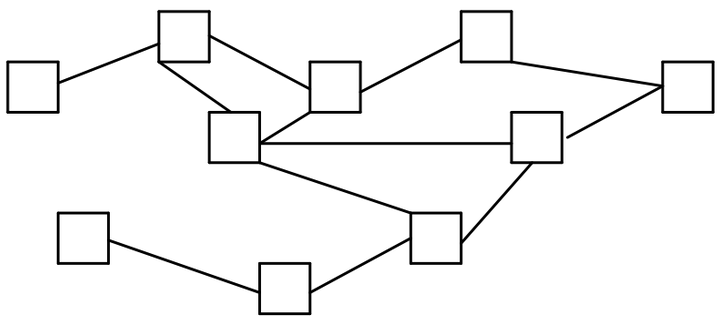
Ten girls live in ten different houses.

The boxes on the diagram represent the girls' houses.

The lines joining the houses show that the two girls from these houses have met.

The girls that have met are:

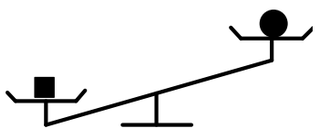
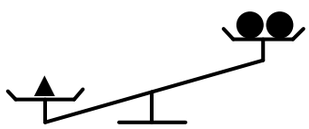
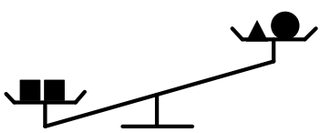
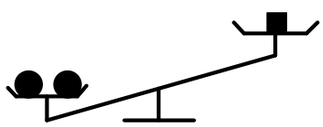
Ella and Alison	Hiba and Justyna
Dee and Fiona	Justyna and Gita
Gita and Bella	Ingrid and Hiba
Alison and Justyna	Hiba and Alison
Crystal and Ingrid	Dee and Hiba
Ingrid and Dee	Crystal and Ella



Put the first letter of the girl's name in the correct house.

Exploration 2

Jane is trying to work out the weights of some shapes. She knows that the weights are whole numbers less than 10kg. The pictures show what Jane finds out. For example, picture (1) shows that the square is heavier than the circle.

(1) 	(2) 
(3) 	(4) 

What are the weights of the shapes?

● ____

■ ____

▲ ____

Exploration 3

Robert has 10 cards, each with a single letter written on it. There are two As, two Bs, two Cs, two Ds and two Fs.
Robert arranges the cards in a certain sequence, so that:
The two As are next to each other;
The two Bs have one letter in between them;
The two Fs have two letters in between them;
The two Ds have three letters in between them;
The two Cs have four letters in between them.
The first two cards are D, then C.
What is the complete sequence in which Robert arranged the cards?

Exploration 4

(1) Alice, Bob, and Charlie make the following statements:
Alice: Bob is telling the truth.
Bob: Alice is telling the truth.
Charlie: Alice is lying.
What are the possible numbers of people telling the truth? Explain your answer.

(2) They now make the following statements:
Alice: Bob and Charlie are both lying.
Bob: Alice is telling the truth or Charlie is lying (or both).
Charlie: Alice and Bob are both telling the truth.
Who is telling the truth and who is lying on this occasion? Explain your answer.

Warm Up

On an island, there are two types of people: Knights and Knaves.
Knights always tell the truth, and Knaves always lie.
You meet two people on the island, a man and a woman.
The man says “We are both Knaves”.
Work out whether each person is a Knight or a Knave.

Exploration 5

There are five people: Alfred, Bryan, Charlie, Daisy and Ellie. Some of them always tell the truth, and the rest always tell lies.
Alfred says, "Daisy never tells the truth".
Bryan says, "Everything Ellie says is false".
Charlie says, "Bryan and I are both liars".
Daisy says, "I only ever tell the truth".
Ellie says, “Daisy is a liar".
Which people are telling the truth, and which are lying?

Exploration 6

Three people called Alf, Beth, and Gemma, sit together in the same room.
One of them always tells the truth.
One of them always tells a lie.
The other one tells truth or lies at random.
In each of the following situations, your task is to determine how each person acts.

(i) Suppose that Alf says "I always tell lies," and Beth says "Yes, that's true, Alf always tells lies."
Who always tells the truth? Who always lies? Briefly explain your answer.

(ii) Suppose instead that Gemma says "Beth always tells the truth," and Beth says "That's wrong."
Who always tells the truth? Who always lies? Briefly explain your answer.

Homework

1 Glen is older than Florence.
Ahmed is older than Zara but younger than Florence.
Oliver is younger than Glen but older than Zara.
Yasmin is younger than Glen.
Ahmed is older than Oliver.
Florence is younger than Yasmin.
Which of these six friends is the youngest?
A. Glen B. Florence C. Ahmed D. Zara E. Oliver
F. Yasmin

2 "David is 11", said Anne.
"I am 13", said David.
"David is older than me", said Meera.
Anne sometimes tells the truth, Meera always tells the truth and David never tells the truth. One of them is 11, another 12 and the other 13.
How old is each person?
Answer: David _____ , Anne _____ , Meera _____

3 Wenlu, Xander, Yasser and Zoe make the following statements:
Wenlu: "Xander is lying."
Xander: "Yasser is lying."
Yasser: "Zoe is telling the truth."
Zoe: "Wenlu is telling the truth."
What are the possible numbers of people telling the truth?
A. 1 or 2 B. 1 or 3 C. 2 D. 2 or 3 E. 3

4 Jack arrives at the Isle of Truth and Lies, where the residents either always tell the truth or always lie. A, B, C, D, and E are all residents of the island.

A says, "B is not telling the truth."

B says, "C is not telling the truth."

C says, "D is lying."

D says, "E and A - one is telling the truth, and the other is lying."

E says, "A is not lying."

How many of these five residents are lying? _____

5 King Arthur had three knights, namely the Truthful Knight, the Lying Knight, and the Random Knight. These three knights each had 1, 2, or 3 attendants. The Truthful Knight always tells the truth, the Lying Knight always lies, and the Random Knight sometimes tells the truth and sometimes lies. Their attendants fully inherit the speaking style of their respective knights.

One day, the three knights engaged in the following conversation:

Knight Lancelot said, "I have two attendants, and they are holding my armour and sword for me."

Knight Gawain said, "I have the largest number of attendants, and Gareth is the Truthful Knight."

Knight Gareth said, "My attendants always tell lies."

How many attendants does Gareth have? _____

Lesson 5

Divisibility Rules & Prime Numbers

Divisibility Rules

Exploration 1

Abdel and Bert sit around a large circular table. Abdel notices there are 24 times as many people sitting between him and Bert going clockwise than going anticlockwise around the table. Given there are more than 2024 people sitting around the table, what is the smallest number of people sitting around that table?

Exploration 2

1 What is the smallest number divisible by each of the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10?

2 Susie has 7 cards. Each has a different number from 0 to 6 written on it:
0, 1, 2, 3, 4, 5, 6
Susie chooses 6 cards and makes a 6-digit number which is a multiple of 2,3,4,5 and 6.
(1) What is the **largest** number Susie could have made?

(2) What is the **smallest** number Susie could have made?

Exploration 3

1 Which of the following integers are divisible by 36?

A. 540 B. 2646 C. 756 D. 1848 E. 3276

2 "A1234B" is a six-digit number with first digit A (not equal to 0) and last digit B.
"A1234B" is a multiple of 9, but not a multiple of 18.
How many different such numbers "A1234B" are there?

Exploration 4

1 (a) Find the smallest integer greater than 100 which is divisible by 17.

(b) How many integers are there between 100 and 1000 which are divisible by 17?

2 Paul chooses a whole number between 100 and 625.
When the number is divided by 5, the remainder is 3.
When the number is divided by 6, the remainder is 4.
When the number is divided by 9, the remainder is 7.
How many possibilities are there for the number Paul has chosen? _____

Prime Numbers

Exploration 5

Some prime numbers can be written as the sum of two square numbers. For example:

$13=3^2+2^2$

$29=5^2+2^2$

Calculate the next three prime numbers after 29 which can be written as the sum of two square numbers.

Exploration 6

Hard question:

How many different prime numbers are factors of 239,400?

Really hard question:

How many different prime numbers are factors of 1,406,790?

Homework

1 A group of children are standing in a circle to play a game. Between two friends, Charlotte and David, there are 14 times as many children when counting clockwise as there are when counting anticlockwise. If there are more than 320 children in the circle, what is the smallest possible number of children? _____

2 What is the smallest number divisible by each of the numbers 12, 14, 16 and 18? _____

3 “A641B” is a five-digit number with first digit A (not equal to 0) and last digit B. Different letters represent different numbers.
“A641B” is a multiple of 12, but not a multiple of 9.
How many different such numbers “A641B” are there? _____

4 Reuben chooses a whole number between 1 and 1000.
When the number is divided by 2, the remainder is 1.
When the number is divided by 3, the remainder is 2.
When the number is divided by 4, the remainder is 3.
When the number is divided by 5, the remainder is 4.
How many possibilities are there for the number Reuben has chosen? _____

5 How many different prime numbers are factors of 291,060? _____

Lesson 6

Combinatorics

Exploration 1

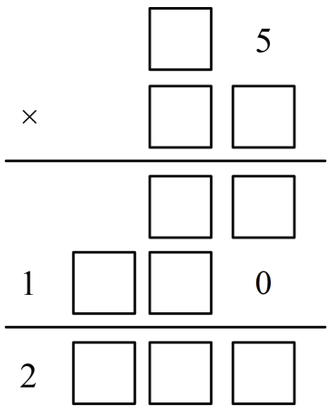
1 A, B and C represent different digits.

$$AB + C = 50$$

$$BC + A = 41$$

What are the values of A, B and C?

2 What are the missing digits in this calculation?



Exploration 2

In the magic square below, the four numbers in each row, column and long diagonal add to the same total.

	15		
8		**3**	**13**
	5	**16**	
14			**7**

Complete the magic square using the extra numbers listed, using each number once only.
Numbers to be added:

1	**2**	**4**	**6**	**9**	**10**	**11**	**12**

Exploration 3

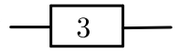
1st January 2000 was a Saturday. What day of the week will 1st January 3000 fall on?

To be a leap year, the year must be a multiple of 4. However, years that are multiples of 100 are NOT leap years unless they are multiples of 400.

So for example, 1900 was not a leap year (multiple of 100, not a multiple of 400) , but 2000 was a leap year (multiple of 400).

Exploration 4

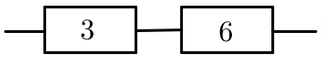
The "Electric Light Organization" makes many electrical components, including Zisters which control the power to any electrical appliance.

This Zister 

has a value of 3,

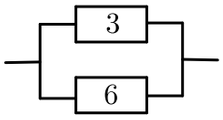
Zisters can be put together in two different ways as follows ,

In an AFFTA, the two values of the Zisters are added, so for example,



has a value of 9, because $3 + 6 = 9$.

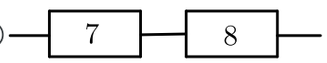
In a NEXTA, the value has to be worked out by adding fractions as follows,

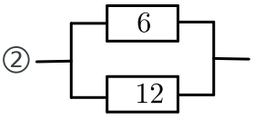


Since $\frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$,

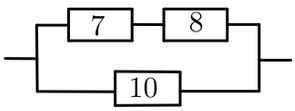
the value of this NEXTA is 2, taken from the answer to the fraction sum.

(1) Work out the values of the following combinations of Zisters in these two questions using the methods in the examples above.

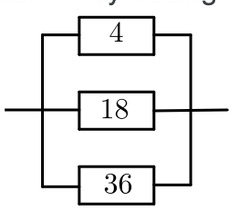
① 

② 

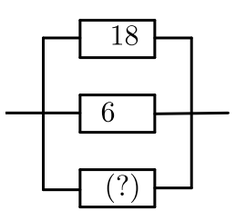
(2) Work out the value of this combination of AFFTA and NEXTA.



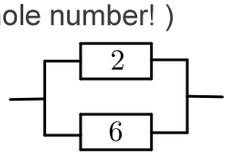
(3) Work out the value of this triple NEXTA by adding three fractions.



(4) If this new triple NEXTA has a value of 3, work out the value of the missing Zister shown with (?)



(5) Finally, here is a special NEXTA. Showing all your working, calculate its value. (it is special because its value is not a whole number!)



Exploration 5

The output of a heater is measured in watts and kilowatts. 1 kilowatt (kW) is equal to 1000 Watts so, for example a 2.6 kW heater produces 2600 Watts.

The Retention Factor (RF) of an insulating layer (which is material that stops heat flowing out) shows what percentage of the heat that reaches the insulating layer is kept in, and also allows you to work out what percentage of the heat escapes from the other side.

The table below gives you some examples of the percentages for certain RFs. Use the patterns in the table to work out the percentages you will need in the questions.

Retention Factor (RF)	Percentage of heat kept in	Percentage of heat that escapes
100	100	0
60	60	40
20	20	80
10	10	90

This means that, for example, with a 5 kW heater, the number of Watts escaping through a 60RF layer is given by

$5000 \times 40 \div 100 = 2000\text{Watts} = 2\text{kW}$

Using this method, work out the answers to the following questions.

(1) With a 3 kW heater, how many Watts of heat escape through a 70RF insulating layer?

(2) With a 2 kW heater, how much heat escapes through two 90RF layers put side by side?

(3) With a 1 kW heater, how many 80RF layers are needed so that the heat that escapes is less than 5 Watts?

(4) A 4 kW heater is in front of three layers with a 50RF layer nearest the heater then a 40RF layer then a 30RF layer. What heat **escapes** between the second (40RF) and third (30RF) layers?

(5) From a 4 kW heater only 108 Watts escape through three **identical** layers. What is the RF of each layer?

Homework

1 In the calculation below, seven digits are missing. What is the sum of the missing digits? _____

$$\begin{array}{r} 2\square \\ \times\ \square 1 \\ \hline \square 4 \\ \square 2\square \\ \hline 7\square\square \end{array}$$

2 In the magic square below, the four numbers in each row, column and long diagonal add to the same total.
Complete the magic square using the extra numbers listed, using each number once only. What is the first number in the fourth row? _____

1	14	7	
8	11		
10			3
		9	6

Numbers to be added:

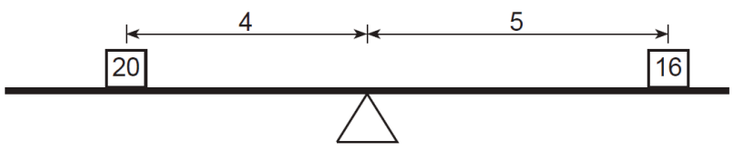
2	4	5	12	13	15	16

3 1st May 2020 was a Friday. What day of the week will 1st May 2200 fall on?

A. Monday B. Tuesday C. Wednesday D. Thursday

E. Friday F. Saturday G. Sunday

4 On a see-saw, the twist of a mass is found by multiplying the mass by the distance of the mass from the balance point.

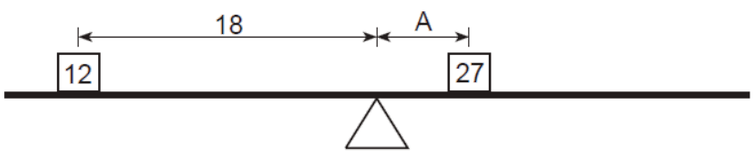


The masses are 20 and 16 and the distances are 4 and 5.

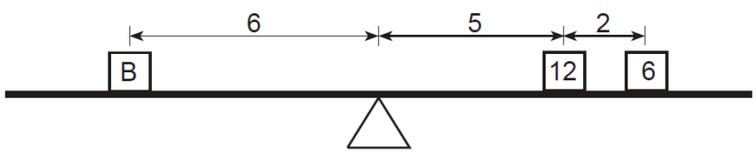
The twist on one side = 20 x 4 = 80. The twist on the other side = 16 x 5 = 80.

So the see-saw is balanced.

(i) Work out the value of A. _____



(ii) Work out the value of B. _____



5 The stopping distance for a car is made up of two parts. The first is the distance travelled by the car while the driver is reacting to something that makes them want to brake. This is called the **Thinking Distance**. The second is the distance travelled by the car while the brakes are applied. This is called the **Braking Distance**. The **Stopping Distance** is given by adding these two distances together.

So **Stopping Distance** = **Thinking Distance** + **Braking Distance**

The table below shows the **Thinking Distance**, in metres, for various speeds in km per hour.

Speed (kmph)	Thinking Distance(m)
40	12
50	15
60	18
70	21
80	24
90	27

The **Braking Distance** in metres is given by the formula

Braking Distance $= \frac{1}{60} \times \text{speed} \times \text{speed}$ or $= \frac{1}{60} \times (\text{speed})^2$.

(1) If the Thinking Distance is 24m, what speed was the car travelling at? _____ kmph

(2) What is the Braking Distance of a car travelling at 90kmph? _____ m

(3) What is the Stopping Distance of a car travelling at 90kmph? _____ m

(4) A driver sees a child start to cross the road 80m in front of his car. What distance would the car be from the child when the driver stopped if he was initially travelling at 60kmph. _____ m

Homework Answers

Lesson 1 Fractions

Homework

1. 1.8; 153; 85
2. 1424.5
3. 225
4. $\frac{3}{5}$
5. 15

Lesson 2 Ratios

Homework

1. 11250
2. 20.8
3. $\frac{75}{103}$
4. 15
5. $\frac{1}{3}$

Lesson 3 Algebra

Homework

1. 58
2. 42
3. 2.40
4. 2
5. 4

Lesson 4 Logic Puzzles

Homework

1. D
2. 12; 13; 11
3. B
4. 2
5. 3

Lesson 5 Divisibility Rules & Prime Numbers

Homework

1. 332
2. 1008
3. 4
4. 16
5. 5

Lesson 6 Combinatorics

Homework

1. 24
2. 15
3. D
4. 8; 17
5. (1) 80 (2) 135 (3) 162 (4) 2

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